

ODIR-5K Dataset Read-Only Audit Report

STATUS: DATASET PARTIALLY SUITABLE

1. Dataset Location

- **Primary Archive Directory:** `ML_project/data/odir5k/ODIR-5K/ODIR-5K/` (and root `data/odir5k/ODIR-5K/ODIR-5K/`)
- **Metadata Spreadsheet:** `ML_project/data/odir5k/ODIR-5K/ODIR-5K/data.xlsx`
- **Training Image Directory:** `ML_project/data/odir5k/ODIR-5K/ODIR-5K/Training Images/`
- **Testing Image Directory:** `ML_project/data/odir5k/ODIR-5K/ODIR-5K/Testing Images/`
- **Preprocessed Image Cache:** `ML_project/data/odir5k/preprocessed_images/` (6,392 images)
- **Consolidated Metadata CSV:** `ML_project/data/odir5k/full_df.csv` (6,392 records)

2. Dataset Structure

The physical directory tree verified locally on disk is structured as follows:

data/odir5k/	
├─ full_df.csv	# Pre-extracted tabular metadata (6,392 rows)
├─ preprocessed_images/	# Preprocessed fundus image cache (6,392 .jpg files)
└─ ODIR-5K/	
└─ ODIR-5K/	
├─ data.xlsx	# Master Ground-Truth Metadata Spreadsheet (1.3 MB)
├─ Training Images/	# 7,000 fundus image files (.jpg)
└─ Testing Images/	# 1,000 fundus image files (.jpg)

- **Folder Structure Summary:**
- `Training Images/` : Subdirectory containing all 7,000 paired training images for 3,500 patients.
- `Testing Images/` : Subdirectory containing 1,000 paired test images for 500 challenge evaluation patients.
- `data.xlsx` : Official Peking University ODIR-5K challenge annotations for the 3,500 training patients.

3. Image Statistics

Metric	Training Images	Testing Images	Preprocessed Cache	Combined Total		
Total Image Files	7,000	1,000	6,392	8,000 raw / 6,392 cached		
File Formats / Extensions	100% .jpg	100% .jpg	100% .jpg	100% .jpg		
Corrupted / Zero-byte Files	0	0	0	0		
Left-Eye Images	3,500 (_left.jpg)	500 (_left.jpg)	3,196 (*_left.jpg)	4,000 raw		
Right-Eye Images	3,500 (_right.jpg)	500 (_right.jpg)	3,196 (*_right.jpg)	4,000 raw		
Filename Pattern Adherence	100% (^\d+_(left\	right)\.jpg\$)	100% (^\d+_(left\	right)\.jpg\$)	100%	100.0% Strict Conformance

- **Image Filename Pattern Verification:**
- Standard training format: 0_left.jpg , 0_right.jpg , 1_left.jpg , 1_right.jpg , ..., 4784_right.jpg .
- Standard testing format: 1000_left.jpg , 1000_right.jpg , 1001_left.jpg , 1001_right.jpg , ..., 4794_right.jpg .
- Exactly 100% of images in both directories adhere to the strict regex _.jpg .

4. Metadata Statistics (data.xlsx)

- **Total Rows:** 3,500 patient records.
- **Total Columns:** 15 columns.
- **All 15 Column Names:**
- ID
- Patient Age
- Patient Sex
- Left-Fundus
- Right-Fundus
- Left-Diagnostic Keywords
- Right-Diagnostic Keywords
- N (Normal)
- D (Diabetes)
- G (Glaucoma)
- C (Cataract)
- A (Age-related Macular Degeneration)
- H (Hypertension)
- M (Pathological Myopia)
- O (Other diseases / abnormalities)

Data Types & Missing Values Audit

Column Name	Inferred Type	Python Data Type	Missing Count	% Missing	Unique Values
ID	Integer ID	int64	0	0.0%	3,500
Patient Age	Discrete Numeric	int64	0	0.0%	76 (range: 1 – 91, mean: 57.85)
Patient Sex	Categorical	object / str	0	0.0%	2 ('Male': 1,885, 'Female': 1,615)
Left-Fundus	Filename String	str	0	0.0%	3,500
Right-Fundus	Filename String	str	0	0.0%	3,500
Left-Diagnostic Keywords	Text / Clinical Annotation	str	0	0.0%	Free text clinical notes
Right-Diagnostic Keywords	Text / Clinical Annotation	str	0	0.0%	Free text clinical notes
N	Binary Indicator	int64	0	0.0%	2 (0 or 1)
D	Binary Indicator	int64	0	0.0%	2 (0 or 1)
G	Binary Indicator	int64	0	0.0%	2 (0 or 1)
C	Binary Indicator	int64	0	0.0%	2 (0 or 1)
A	Binary Indicator	int64	0	0.0%	2 (0 or 1)
H	Binary Indicator	int64	0	0.0%	2 (0 or 1)
M	Binary Indicator	int64	0	0.0%	2 (0 or 1)
O	Binary Indicator	int64	0	0.0%	2 (0 or 1)

First 5 Rows of `data.xlsx`

```
Row 0:
  ID: 0 | Age: 69 | Sex: Female | Left-Fundus: 0_left.jpg | Right-Fundus: 0_right.jpg
  Keywords: Left="cataract" | Right="normal fundus"
  Labels: N=0, D=0, G=0, C=1, A=0, H=0, M=0, O=0

Row 1:
  ID: 1 | Age: 57 | Sex: Male | Left-Fundus: 1_left.jpg | Right-Fundus: 1_right.jpg
  Keywords: Left="normal fundus" | Right="normal fundus"
  Labels: N=1, D=0, G=0, C=0, A=0, H=0, M=0, O=0

Row 2:
  ID: 2 | Age: 42 | Sex: Male | Left-Fundus: 2_left.jpg | Right-Fundus: 2_right.jpg
  Keywords: Left="laser spots,..." | Right="moderate non proliferative retinopathy"
  Labels: N=0, D=1, G=0, C=0, A=0, H=0, M=0, O=1

Row 3:
  ID: 3 | Age: 66 | Sex: Male | Left-Fundus: 3_left.jpg | Right-Fundus: 3_right.jpg
  Keywords: Left="dry age-related macular degeneration" | Right="dry age-related macular degeneration"
  Labels: N=0, D=0, G=0, C=0, A=1, H=0, M=0, O=1

Row 4:
  ID: 4 | Age: 53 | Sex: Male | Left-Fundus: 4_left.jpg | Right-Fundus: 4_right.jpg
  Keywords: Left="macular epiretinal membrane" | Right="mild nonproliferative retinopathy"
  Labels: N=0, D=1, G=0, C=0, A=0, H=0, M=0, O=1
```

5. Patient ID Analysis

- **Unique Patient IDs:** Exactly 3,500 unique patient IDs in `data.xlsx`.
- **ID Numbering Scheme:** Non-contiguous integer IDs ranging from `0` to `4784` in the training set; the remaining 500 IDs up to `4794` are allocated to the testing set.
- **Row-to-ID Relationship:** Each row represents exactly one unique human subject.
- **ID Integrity:** No duplicate IDs, no null IDs, and no synthetic/fabricated IDs exist in the source metadata.

6. Image-to-Patient Linkage Audit

Linkage Dimension	Audited Count	Percentage
Total Metadata Patient Records	3,500	100.0%
Valid Left-Eye Image Matches (Training Images/)	3,500	100.0%
Valid Right-Eye Image Matches (Training Images/)	3,500	100.0%
Missing Left-Eye Images	0	0.0%
Missing Right-Eye Images	0	0.0%
Bilateral Paired Image Matches (Both Left & Right Exist)	3,500	100.0%

- **Linkage Mechanism:**
- Linkage is **explicit and deterministic** via filename string references in `Left-Fundus` and `Right-Fundus` columns, which match physical files on disk:
- Patient `0` $\rightarrow$ `Left-Fundus = "0_left.jpg"` , `Right-Fundus = "0_right.jpg"`.
- Patient `1005` $\rightarrow$ `Left-Fundus = "1005_left.jpg"` , `Right-Fundus = "1005_right.jpg"`.

- Linkage is **NOT** inferred from row position or file sorting order.
 - Zero missing files or broken references were detected across the entire 3,500-patient training cohort.
-

7. Clinical Feature Analysis

A systematic audit of tabular clinical features was conducted to assess feature richness:

Available Features (Genuinely Present in ODIR-5K)

- **Patient Age** : Continuous integer (1 to 91 years; Mean: 57.85 $\pm$ 11.72 years).
- **Patient Sex** : Categorical ('Male' = 1,885, 'Female' = 1,615).
- **Diagnostic Keywords** : Free-text clinical diagnosis per eye.

Not Available Features (Completely Absent from ODIR-5K)

- **Glucose** / Fasting Blood Sugar: **NOT AVAILABLE**
 - **HbA1c** (Glycated Hemoglobin): **NOT AVAILABLE**
 - **Creatinine** : **NOT AVAILABLE**
 - **BUN** (Blood Urea Nitrogen): **NOT AVAILABLE**
 - **Blood Pressure** (Systolic BP / Diastolic BP): **NOT AVAILABLE**
 - **HDL Cholesterol** : **NOT AVAILABLE**
 - **Total Cholesterol** : **NOT AVAILABLE**
 - **BMI** (Body Mass Index): **NOT AVAILABLE**
 - **Waist Circumference** : **NOT AVAILABLE**
 - **eGFR** (Estimated Glomerular Filtration Rate): **NOT AVAILABLE**
-

8. Disease Label Analysis

The ODIR-5K dataset provides multi-label diagnostic ground truth across 8 standard ophthalmic disease categories:

Code	Disease Category Name	Positive Cases	Prevalence (%)	Clinical Meaning & Pathology
N	Normal	1,140	32.57%	Normal ocular fundus without signs of retinopathy or maculopathy.
D	Diabetes / Diabetic Retinopathy	1,128	32.23%	Retinal vascular damage from diabetes: microaneurysms, hemorrhages, hard/soft exudates, neovascularization.
G	Glaucoma	215	6.14%	Glaucomatous optic neuropathy characterized by enlarged cup-to-disc ratio and neuroretinal rim thinning.
C	Cataract	212	6.06%	Crystalline lens opacification causing hazy/blurred fundus photography.
A	Age-related Macular Degeneration (AMD)	164	4.69%	Macular drusen, geographic atrophy (dry AMD), or choroidal neovascular membrane (wet AMD).
H	Hypertension / Hypertensive Retinopathy	103	2.94%	Retinal vascular changes from chronic high BP: arteriolar narrowing, arteriovenous nicking, flame hemorrhages.
M	Pathological Myopia	174	4.97%	High myopia degenerative changes: posterior staphyloma, chorioretinal atrophy, lacquer cracks.
O	Other Diseases / Abnormalities	979	27.97%	Other ocular conditions: epiretinal membrane, retinal vein occlusion (RVO), macular hole, retinal detachment, laser scars.

Note: Total positive labels sum to 4,115 across 3,500 patients because ODIR is a multi-label classification dataset where patients can have co-occurring ocular conditions (e.g., Diabetic Retinopathy + Hypertension or Cataract + AMD).

9. NHANES Compatibility

- **Direct Linkage Possibility: STRICTLY NO.**
- **Reasoning:**
- **Cohorts:** NHANES is a continuous United States population health survey conducted by the US CDC / NCHS. ODIR-5K is an ophthalmic hospital challenge cohort collected across multiple medical centers in China (Peking University, Shangong Medical Technology).
- **Identifiers:** There is zero overlap in patient identification numbers ([SEQN](#) in NHANES vs [ID](#) in ODIR-5K).
- **Scientific Integrity:** Matching ODIR fundus images with NHANES clinical records based on age, sex, row order, or nearest-neighbor similarity is scientifically fraudulent and constitutes artificial data leakage.

10. PTB-XL Compatibility

- **Direct Linkage Possibility: STRICTLY NO.**
- **Reasoning:**
- PTB-XL is an Austrian clinical electrocardiology database (12-lead ECG waveforms collected by Physikalisch-Technische Bundesanstalt).
- ODIR-5K is a Chinese retinal fundus photography database.
- The patient cohorts are completely disjoint with zero shared biological or clinical identifiers.

11. Existing Architecture Compatibility

The existing project codebase contains modular components designed for multimodal processing:

1. **ClinicalEncoder** (`app/models/clinical_encoder.py`):
 2. Multi-layer perceptron mapping tabular clinical features ($d_{\text{in}} \rightarrow [128, 64] \rightarrow 64$).
 3. Can encode ODIR demographics (`Age` and `Sex` , $d_{\text{in}}=2$) into a 64-dimensional clinical embedding.
 4. **ImageEncoder** (`app/models/image_encoder.py`):
 5. CNN backbone (ResNet-18, ResNet-50, or EfficientNet-B0) with projection head mapping $3 \times 224 \times 224$ images into 64-dimensional embeddings.
 6. Fully compatible with ODIR left and right retinal fundus images.
 7. **MultimodalFusion** (`app/models/multimodal_model.py`):
 8. Concatenates clinical and image embeddings ($64 + 64 = 128 \rightarrow 256 \rightarrow 128$) to create unified multimodal representations.
 9. **PatientGraphBuilder** (`app/graph/patient_graph.py`):
 10. Builds k -NN patient similarity graphs using cosine similarity over multimodal embeddings, yielding edge indices and similarity weights.
 11. **MultiDiseaseGNN** (`app/models/gnn_model.py`):
 12. Message-passing neural network with multi-head classification layers (`heads = ModuleDict`).
 13. Can be configured with 8 binary heads corresponding to ODIR targets (`N` , `D` , `G` , `C` , `A` , `H` , `M` , `O`).
 14. **ImageGradCAM** (`app/explainability/image_xai.py`):
 15. Computes activation heatmaps over CNN feature maps, highlighting retinal lesions (microaneurysms, hemorrhages, cup-to-disc ratio changes, drusen).
-

12. Multimodal Suitability

- **Bilateral Ocular Multimodal Modeling:** ODIR provides genuine paired data for multimodal representation learning:
$$\text{ODIR Patient} = \{ \text{Left Fundus Image}, \text{Right Fundus Image}, \text{Age}, \text{Sex}, \text{8-Class Diagnostic Target} \}$$
 - **Multi-View Fundus Integration:** Left and right eyes can be processed via dual-stream `ImageEncoder` or averaged feature pooling to capture bilateral systemic manifestations (e.g., bilateral diabetic retinopathy vs unilateral retinal vein occlusion).
 - **Patient Similarity Graph:** A k -NN graph over fused fundus+demographic embeddings can facilitate transductive and inductive GNN message passing for rare ocular pathology detection (e.g., Pathological Myopia or Hypertensive Retinopathy).
-

13. Limitations

1. **Absence of Laboratory Biomarkers:** ODIR does **not** contain blood glucose, HbA1c, lipid profiles, blood pressure measurements, or renal markers.
 2. **Cannot Replace NHANES:** ODIR **cannot** substitute or replace the existing NHANES clinical model for cardiometabolic risk prediction (diabetes, coronary heart disease, chronic kidney disease).
 3. **Domain Specificity:** ODIR predictions are strictly ophthalmic (ocular manifestations and systemic diseases visible in retinal microvasculature, such as Diabetic Retinopathy and Hypertensive Retinopathy), rather than general cardiometabolic panels.
-

14. Final Dataset Classification & Recommended Next Steps

Final Classification

B. Image + limited patient-level metadata dataset

- **Justification:** ODIR-5K contains 8,000 real medical images genuinely paired with 3,500 real human patients, complete with bilateral fundus imaging, age, sex, and expert multi-disease ground truth labels. However, it lacks comprehensive laboratory/clinical chemistry panels, restricting its tabular modality to demographic features.

Recommended Next Step

Implement a Standalone Ophthalmic Multimodal Branch (`ODIR Branch`) without modifying or merging NHANES/PTB-XL branches:

1. **Retain Existing Independent Branches:**
2. **Branch 1 (Cardiometabolic Tabular CDS):** NHANES dataset $\rightarrow$ 10 clinical lab features $\rightarrow$ `ClinicalEncoder` $\rightarrow$ `MultiDiseaseGNN` (Diabetes, Heart Disease, CKD).
3. **Branch 2 (Electrophysiological Multimodal CDS):** PTB-XL dataset $\rightarrow$ 12-lead ECG waveforms + 4 demographics $\rightarrow$ `ECGEncoder` + `ClinicalEncoder` $\rightarrow$ `MultimodalFusion` $\rightarrow$ `MultiDiseaseGNN`.
4. **Add New Branch 3 (Ophthalmic Multimodal CDS):**
5. ODIR-5K dataset $\rightarrow$ Bilateral Fundus Images + Age & Sex $\rightarrow$ `ImageEncoder` (ResNet-18) + `ClinicalEncoder` $\rightarrow$ `MultimodalFusion` $\rightarrow$ `PatientGraphBuilder` $\rightarrow$ `MultiDiseaseGNN` (8 Heads: N, D, G, C, A, H, M, O) $\rightarrow$ `ImageGradCAM` retinal heatmap explainability.
6. **Tri-Modal Clinical Dashboard:**
7. Expose all three specialized clinical decision support modalities as distinct triage workflows within the system.